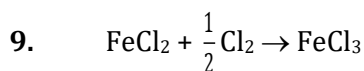


EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Oxidation and Reduction

- Consider the redox reaction $2S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + 2I^-$:
 (A) $2S_2O_3^{2-}$ gets oxidised to $S_4O_6^{2-}$ (B) I_2 gets reduced to I^-
 (C) I_2 gets oxidised to I^- (D) Both (A) & (B)
 CRX096
- The reaction $H_2S + H_2O_2 \rightarrow S + 2H_2O$ shows :
 (A) Oxidation of H_2O_2 (B) Reduction of H_2O_2
 (C) Acidic nature of H_2O_2 (D) Alkaline nature of H_2O_2
 CRX086
- If in a chemical reaction, Fe^{2+} is converted into Fe^{3+} , then Fe^{2+} :
 (A) Gains one electron and gets reduced
 (B) Looses one electron and gets oxidized
 (C) Gains one proton and gets oxidised
 (D) Looses one proton and gets reduced
 CRX087
- In the reaction, $ZnO + C \rightarrow Zn + CO$:
 (A) ZnO is reduced (B) C is oxidised
 (C) Above reaction is a redox reaction (D) All of above
 CRX101
- $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$
 In above reaction :
 (A) both CH_4 and O_2 are oxidised (B) both CH_4 and O_2 are reduced
 (C) CH_4 is oxidised and O_2 is reduced (D) CH_4 is reduced and O_2 is oxidized
 CRX102
- In the reaction, $Zn + CuCl_2 \rightarrow ZnCl_2 + Cu$:
 (A) Zn is getting reduced (B) $CuCl_2$ is getting oxidized
 (C) Both (A) and (B) are correct (D) None of (A) and (B) are correct
 CRX103
- In the reaction, $Mg + F_2 \rightarrow MgF_2$, correct statement(s) is/are :
 (A) Both oxidation and reduction take place in above reaction
 (B) Mg is oxidized
 (C) F_2 is Reduced
 (D) All of the above
 CRX104
- In the reaction, $3Mg + N_2 \rightarrow Mg_3N_2$, incorrect statement is :
 (A) Mg is reduced (B) Mg is oxidised
 (C) N_2 is oxidised (D) Both (A) and (C)
 CRX105



In above reaction :

- (A) Fe is oxidised (B) Cl_2 is reduced
(C) both (A) and (B) are true (D) both (A) and (B) are false

CRX106

Oxidation Number

10. Which statement is incorrect :

- (A) Oxidation number of oxygen is -2 in most of its compounds.
(B) Oxidation number of oxygen is +2 in Oxygen difluoride.
(C) Oxidation number of oxygen is $-\frac{1}{2}$ in superoxides.
(D) Oxidation number of oxygen is +1 in peroxides.

CRX088

11. Identify the correct statement :

- (A) Halogens always have -1 oxidation state in their compounds.
(B) Oxidation number can be zero, integer or fractional.
(C) In OF_2 , the oxidation number of F is +1.
(D) Hydrogen always has + 1 oxidation number in its compounds.

CRX089

12. Among the following compounds given below, what is the sum of the oxidation states of all underlined elements?



- (A) 8 (B) 9 (C) 10 (D) 12

CRX090

13. The oxidation number of Phosphorus in $\text{Mg}_2\text{P}_2\text{O}_7$ is :

- (A) + 3 (B) + 2 (C) + 5 (D) - 3

CRX091

14. The oxidation number of sulphur in S_8 & S_2F_2 respectively are :

- (A) 0, +1 (B) +2, +1 (C) 0, -1 (D) -2, +1

CRX092

15. Manganese achieves its maximum oxidation state in which of these compounds :

- (A) MnO_2 (B) Mn_3O_4 (C) KMnO_4 (D) K_2MnO_4

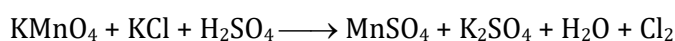
CRX093

16. The incorrect order of decreasing oxidation number of S in compounds is :

- (A) $\text{H}_2\text{S}_2\text{O}_7 > \text{Na}_2\text{S}_4\text{O}_6 > \text{Na}_2\text{S}_2\text{O}_3 > \text{S}_8$ (B) $\text{H}_2\text{SO}_5 > \text{H}_2\text{SO}_3 > \text{SCl}_2 > \text{H}_2\text{S}$
(C) $\text{SO}_3 > \text{SO}_2 > \text{S}_8 > \text{H}_2\text{S}$ (D) $\text{H}_2\text{SO}_4 > \text{SO}_2 > \text{H}_2\text{S} > \text{H}_2\text{S}_2\text{O}_8$

CRX109

17. Identify the oxidant in the following reaction :



- (A) KMnO_4 (B) KCl (C) H_2SO_4 (D) H_2O

CRX110

Redox and Equivalent Concept

18. Identify the reductant in the following reaction :
 $\text{FeCl}_2 + \text{H}_2\text{O}_2 + \text{HCl} \longrightarrow \text{FeCl}_3 + \text{H}_2\text{O}$
 (A) FeCl_2 (B) H_2O_2 (C) HCl (D) $\text{Cl}^\ominus$ CRX111
19. H_2O_2 acts as a oxidising agent in :
 (A) $\text{FeCl}_2 + \text{HCl} + \text{H}_2\text{O}_2 \longrightarrow \text{FeCl}_3 + \text{H}_2\text{O}$ (B) $\text{HI} + \text{H}_2\text{O}_2 \longrightarrow \text{I}_2 + \text{H}_2\text{O}$
 (C) $\text{H}_2\text{SO}_3 + \text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$ (D) All of these CRX112
20. (A) $\text{S}^{2\ominus} + 4\text{H}_2\text{O}_2 \longrightarrow \text{SO}_4^{2\ominus} + 4\text{H}_2\text{O}$
 (B) $\text{Cl}_2 + \text{H}_2\text{O}_2 \longrightarrow 2\text{HCl} + \text{O}_2$
 The true statement regarding the above reactions is :
 (A) H_2O_2 acts as reductant in both the reactions.
 (B) H_2O_2 acts as oxidant in reaction (A) and reductant in reaction (B).
 (C) H_2O_2 acts as an oxidant in both the reactions.
 (D) H_2O_2 acts as reductant in reaction (A) and oxidant in reaction (B). CRX113

Types of Redox Reactions

21. Which of the following is/are decomposition reaction(s) ?
 (A) $2\text{HgO} \longrightarrow 2\text{Hg} + \text{O}_2$ (B) $\text{Fe}_2(\text{SO}_4)_3 \longrightarrow \text{Fe}_2\text{O}_3 + \text{SO}_3$
 (C) $2\text{H}_2\text{O} \longrightarrow 2\text{H}_2 + \text{O}_2$ (D) All of these CRX097
22. Which is a displacement redox reaction :
 (A) $\text{Cr}_2(\text{SO}_4)_3 + 6\text{KOH} \rightarrow 2\text{Cr}(\text{OH})_3 + 3\text{K}_2\text{SO}_4$ (B) $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$
 (C) $\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NH}_3 + \text{NaCl} + \text{H}_2\text{O}$ (D) $2\text{AgNO}_3 + \text{Cu} \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{Ag}$ CRX098
23. Which of the following is redox reaction?
 (A) $\text{NaOH} + \text{HCl} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$ (B) $\text{AgNO}_3 + \text{KI} \longrightarrow \text{AgI} + \text{KNO}_3$
 (C) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$ (D) $\text{SnCl}_2 + \text{HgCl}_2 \longrightarrow \text{SnCl}_4 + \text{Hg}$ CRX114
24. Which of the following is a combination redox reaction ?
 (A) $\text{CaO} + \text{CO}_2 \longrightarrow \text{CaCO}_3$ (B) $\text{H}_2 + \text{Cl}_2 \longrightarrow 2\text{HCl}$
 (C) $\text{CaO} + 2\text{HCl} \longrightarrow \text{CaCl}_2 + \text{H}_2\text{O}$ (D) $\text{NaOH} + \text{HCl} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$ CRX115

Disproportionation, Comproportionation, Intramolecular Redox

25. Which reaction does not represent auto redox or disproportionation :
 (A) $\text{Cl}_2 + \text{OH}^\ominus \longrightarrow \text{Cl}^\ominus + \text{ClO}_3^\ominus + \text{H}_2\text{O}$ (B) $2\text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{O}_2$
 (C) $2\text{Cu}^\oplus \longrightarrow \text{Cu}^{2\oplus} + \text{Cu}$ (D) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \longrightarrow \text{N}_2 + \text{Cr}_2\text{O}_3 + 4\text{H}_2\text{O}$ CRX099

26. Which of the following is not a comproportionation redox reaction?
 (A) $\text{Na}_2\text{SO}_3 + \text{S} \rightarrow \text{Na}_2\text{S}_2\text{O}_3$ (B) $2\text{BaO} + \text{O}_2 \rightarrow 2\text{BaO}_2$
 (C) $\text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{HCO}_3)_2$ (D) $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 2\text{H}_2\text{O} + 3\text{S}$

CRX100

27. Which of the following is example of disproportionation reaction :
 (A) $\text{HgO} \longrightarrow \text{Hg} + \text{O}_2$ (B) $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$
 (C) $\text{KClO}_3 \longrightarrow \text{KClO}_4 + \text{KCl}$ (D) $\text{AgNO}_3 + \text{KI} \longrightarrow \text{AgI} + \text{KNO}_3$

CRX116

Equivalent weight & n-factor

28. The equivalent weight of metal (Atomic Mass = 24 u), if 40 g metal oxide contains 60% metal by mass, is :
 (A) 8 (B) 12 (C) 16 (D) 24

CRX118

29. Equivalent mass of H_3PO_4 in below reaction is :
 $\text{H}_3\text{PO}_4 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{HPO}_4 + 2\text{H}_2\text{O}$
 (A) 98 (B) 49 (C) 32.67 (D) 196

CRX119

30. In acidic medium, potassium dichromate acts as an oxidant according to the equation :
 $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$
 What is the equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ (mol. wt. = M) ?
 (A) M (B) M/2 (C) M/3 (D) M/6

CRX120

31. The equivalent weight of $\text{Na}_2\text{S}_2\text{O}_3$ (Mol. wt. = M) in the reaction, $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \longrightarrow \text{Na}_2\text{S}_4\text{O}_6 + 2\text{NaI}$ is :
 (A) M/4 (B) M/3 (C) M/2 (D) M

CRX121

32. Equivalent mass of oxidising agent in the reaction, $\text{SO}_2 + 2\text{H}_2\text{S} \longrightarrow 3\text{S} + 2\text{H}_2\text{O}$ is :
 (A) 32 (B) 64 (C) 16 (D) 8

CRX122

33. In the change, $3\text{Fe} + 4\text{H}_2\text{O} \longrightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$
 if the atomic weight of iron is 56 u, then its equivalent weight will be :
 (A) 42 (B) 21 (C) 63 (D) 84

CRX123

34. Equivalent weight of FeC_2O_4 in the change : $\text{FeC}_2\text{O}_4 \longrightarrow \text{Fe}^{3+} + \text{CO}_2$ is : (mol. wt. = M)
 (A) M/3 (B) M/6 (C) M/2 (D) M/1

CRX124

35. The valence factor of Cu_2S in below change is :
 $\text{Cu}_2\text{S} \longrightarrow \text{Cu}^{2+} + \text{SO}_4^{2-}$
 (A) 2 (B) 4 (C) 8 (D) 10

CRX125

36. In the reaction, $\text{VO} + \text{Fe}_2\text{O}_3 \rightarrow \text{FeO} + \text{V}_2\text{O}_5$, the eq. wt. of V_2O_5 is equal to its :
 (A) Mol. wt. (B) Mol. wt./8 (C) Mol. wt./6 (D) Mol. wt./2

CRX128

Redox and Equivalent Concept

37. In the reaction : $A^{n_2\ominus} + xe^{\ominus} \longrightarrow A^{n_1\ominus}$, here x will be
 (A) $n_1 + n_2$ (B) $n_2 - n_1$ (C) $n_1 - n_2$ (D) $n_1 \cdot n_2$ CRX129
38. In a reaction 4 mole of electrons are transferred to one mole of HNO_3 when it acts as an oxidant. The possible reduction product is :
 (A) $(1/2)$ mole N_2 (B) $(1/2)$ mole N_2O (C) 1 mole of NO_2 (D) 1 mole NH_3 CRX130
39. The equivalent weight of MnSO_4 is half of its molecular weight when it is converted to :-
 (A) Mn_2O_3 (B) MnO_2 (C) $\text{MnO}_4^{\ominus}$ (D) $\text{MnO}_4^{2\ominus}$ CRX131
40. $\text{Cr}_2\text{O}_7^{2\ominus} + \text{I}^{\ominus} + \text{H}^{\oplus} \longrightarrow \text{Cr}^{3\oplus} + \text{I}_2 + \text{H}_2\text{O}$
 The equivalent weight of the reductant in the above equation is :-(At. wt. of Cr = 52, I = 127)
 (A) 26 (B) 127 (C) 63.5 (D) 10.4 CRX132

Balancing of Redox Reaction

41. In the reaction, $x\text{HI} + y\text{HNO}_3 \longrightarrow \text{NO} + \text{I}_2 + \text{H}_2\text{O}$:
 (A) $x = 3, y = 2$ (B) $x = 2, y = 3$ (C) $x = 6, y = 2$ (D) $x = 6, y = 1$ CRX251
42. For the redox reaction, $\text{Cr}_2\text{O}_7^{2\ominus} + \text{C}_2\text{O}_4^{2\ominus} + \text{H}^{\oplus} \longrightarrow \text{Cr}^{3\oplus} + \text{CO}_2 + \text{H}_2\text{O}$,
 the correct whole number stoichiometric coefficients of $\text{Cr}_2\text{O}_7^{2\ominus}$, $\text{C}_2\text{O}_4^{2\ominus}$ and $\text{H}^{\oplus}$ are respectively :
 (A) 1, 3, 14 (B) 14, 3, 1 (C) 3, 14, 1 (D) 1, 14, 3 CRX252
43. For the redox reaction $x\text{P}_4 + y\text{HNO}_3 \longrightarrow \text{H}_3\text{PO}_4 + \text{NO}_2 + \text{H}_2\text{O}$, upon balancing with whole number coefficients :
 (A) $x = 1, y = 5$ (B) $x = 2, y = 10$ (C) $x = 1, y = 20$ (D) $x = 1, y = 15$ CRX253
44. $x\text{NO}_3^{\ominus} + y\text{I}^{\ominus} + z\text{H}^{\oplus} \longrightarrow 2\text{NO} + 3\text{I}_2 + 4\text{H}_2\text{O}$; x, y, z respectively in the above equation are :
 (A) 2, 6, 8 (B) 1, 6, 4 (C) 0, 6, 8 (D) 2, 3, 4 CRX254
45. In the reaction $\text{X}^{\ominus} + \text{XO}_3^{\ominus} + \text{H}^{\oplus} \longrightarrow \text{X}_2 + \text{H}_2\text{O}$, the molar ratio in which $\text{X}^{\ominus}$ and $\text{XO}_3^{\ominus}$ react is :
 (A) 1 : 5 (B) 5 : 1 (C) 2 : 3 (D) 3 : 2 CRX255
46. Balance the following equation and choose the quantity which is the sum of the coefficients of reactants and products :
 $\text{..... KMnO}_4 + \text{..... H}_2\text{O}_2 + \text{..... H}_2\text{SO}_4 \longrightarrow \text{MnSO}_4 + \text{..... O}_2 + \text{..... H}_2\text{O} + \text{..... K}_2\text{SO}_4$
 (A) 26 (B) 23 (C) 28 (D) 22 CRX257
47. $\text{CN}^{\ominus}$ is oxidised by $\text{NO}_3^{\ominus}$ in presence of acid :
 $a\text{CN}^{\ominus} + b\text{NO}_3^{\ominus} + c\text{H}^{\oplus} \longrightarrow (a+b)\text{NO} + a\text{CO}_2 + \frac{c}{2}\text{H}_2\text{O}$
 What are the whole number values of a, b, c in that order :
 (A) 3, 7, 7 (B) 3, 10, 7 (C) 3, 10, 10 (D) 3, 7, 10 CRX259

48. The number of electrons to balance the following equation, $\text{NO}_3^- + 4\text{H}^+ + e^- \rightarrow 2\text{H}_2\text{O} + \text{NO}$ is :
 (A) 5 (B) 4 (C) 3 (D) 2 CRX249
49. Number of moles of electrons taken up when 1 mole of NO_3^- ions is reduced to 1 mole of NH_2OH is
 (A) 2 (B) 4 (C) 5 (D) 6 CRX250
50. When arsenic sulphide is boiled with NaOH, sodium arsenite and sodium thioarsenite are formed according to reaction :

$$x\text{As}_2\text{S}_3 + y\text{NaOH} \longrightarrow x\text{Na}_3\text{AsO}_3 + x\text{Na}_3\text{AsS}_3 + \frac{y}{2}\text{H}_2\text{O}$$
 What are the values of x and y?
 (A) 1, 6 (B) 2, 8 (C) 2, 6 (D) 1, 4 CRX258

Equivalent Concept

51. The mass of oxalic acid crystals ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) that would contain its 0.01 equivalents is :
 (A) 4.5 g (B) 6.3 g (C) 0.63 g (D) 0.45 g CRX127
52. The number of milliequivalents in 1.58 g of KMnO_4 in a reaction where it gets reduced to Mn^{2+} is :
 (At. wt. : K = 39 u, Mn = 55 u)
 (A) 50 (B) 2 (C) 5 (D) 10 CRX192
53. What volume in litre of 0.1 N $\text{K}_2\text{Cr}_2\text{O}_7$ solution should be taken to have 1 equivalent of $\text{K}_2\text{Cr}_2\text{O}_7$ in a reaction where it is reduced to Cr^{3+} ?
 (A) 1 L (B) 0.1 L (C) 100 L (D) 10 L CRX193
54. The milliequivalents of H_3PO_4 in 120 mL of 1.5 M H_3PO_4 solution in a reaction where it undergoes complete neutralization is :
 (A) 180 (B) 360 (C) 540 (D) 60 CRX194
55. In the redox reaction, $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \longrightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$, 20 mL of 0.1 M KMnO_4 react quantitatively with :
 (A) 20 mL of 0.1 M oxalate (B) 40 mL of 0.1 M oxalate
 (C) 50 mL of 0.25 M oxalate (D) 50 mL of 0.1 M oxalate CRX195
56. One mole of acidified $\text{K}_2\text{Cr}_2\text{O}_7$ on reaction with sufficient KI will liberate mole(s) of I_2 , along with Cr^{3+} .
 (A) 6 (B) 1 (C) 7 (D) 3 CRX196
57. 1 mol each of H_3PO_2 , H_3PO_3 and H_3PO_4 will neutralise respectively x mol of NaOH, y mol of $\text{Ca}(\text{OH})_2$ and z mol of $\text{Al}(\text{OH})_3$ (assuming all as strong electrolytes). x, y, z are in the ratio :
 (A) 3 : 1.5 : 1 (B) 1 : 2 : 3 (C) 3 : 2 : 1 (D) 1 : 1 : 1 CRX199

Redox and Equivalent Concept

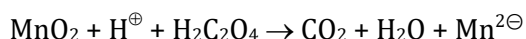
58. 125 mL of 63% (w/v) $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ solution is made to react with 125 mL of a 40% (w/v) NaOH solution. The resulting solution is: (ignoring hydrolysis of ions)
 (A) neutral (B) acidic (C) strongly acidic (D) alkaline

CRX200

59. A substance which participates readily in both acid-base and oxidation-reduction reactions is :
 (A) Na_2CO_3 (B) KOH (C) KMnO_4 (D) $\text{H}_2\text{C}_2\text{O}_4$

CRX201

60. What mass of MnO_2 is reduced by 35 ml of 0.16 N oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$) in acidic solution? The skeleton equation is :-



- (A) 8.7 g (B) 0.24 g (C) 0.84 g (D) 43.5 g

CRX203

Redox Titration

61. Which of the following solutions will exactly oxidize 25 mL of an acidic solution of 0.1 M iron (II) oxalate:

- (A) 25 mL of 0.1 M $\text{K}_2\text{Cr}_2\text{O}_7$ (B) 25 mL of 0.2 M $\text{K}_2\text{Cr}_2\text{O}_7$
 (C) 25 mL of 0.6 M $\text{K}_2\text{Cr}_2\text{O}_7$ (D) 12.5 mL of 0.1 M $\text{K}_2\text{Cr}_2\text{O}_7$

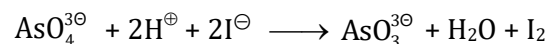
CRX224

62. How many litres of Cl_2 gas at NTP (1 ATM & 273K) will be liberated by the oxidation of NaCl with 15.8 g KMnO_4 in acidic medium ? (Atomic weight : Mn = 55 u and K = 39 u)

- (A) 5.6 (B) 11.2 (C) 56 (D) 1.12

CRX225

63. 1.04 g of Na_3AsO_4 is boiled with excess of solid KI in presence of strong HCl. The iodine evolved is absorbed in KI solution and titrated against 0.2 N hypo solution. Assuming the reaction to be



calculate the volume of hypo consumed. [Atomic weight of As = 75 u]

- (A) 100 mL (B) 50 mL (C) 25 mL (D) 5 mL

CRX226

64. Dichloroacetic acid (CHCl_2COOH) is oxidised to CO_2 , H_2O and Cl_2 by 600 m.eq. of an oxidising agent. Same amount of acid can neutralize how many moles of ammonia to form ammonium dichloroacetate :

- (A) 0.0167 (B) 0.1 (C) 0.3 (D) 0.6

CRX227

65. If 9.1 g of V_2O_5 is dissolved in acid and is reduced to $\text{V}^{2\oplus}$ by zinc metal, how many mole of I_2 could be reduced by the resulting solution, if it is further oxidised to $\text{VO}^{2\oplus}$ ions :

[Assume no change in state of $\text{Zn}^{2\oplus}$ ions] (Atomic masses : V = 51 u, O = 16 u, I = 127 u)

- (A) 0.1 (B) 0.2 (C) 0.05 (D) 0.4

CRX228

66. How many gram of KMnO_4 are contained in 4 litre of 0.05 N solution. The KMnO_4 is to be used as an oxidant in acidic medium ?

- (A) 1.58 g (B) 15.8 g (C) 6.32 g (D) 31.6 g

CRX236

67. 40 ml, 0.5 M - $\text{Na}_2\text{S}_2\text{O}_3$ solution is required for complete reaction with 25 ml I_2 solution. The molarity of I_2 solution is -
(A) 0.4M (B) 0.8 M (C) 0.25 M (D) 0.16 M

CRX237

68. 25 ml acidified of $\text{K}_2\text{Cr}_2\text{O}_7$ solution is diluted to 200 ml. If 40 ml of the diluted solution exactly required 15 ml of 0.2 N - KI solution for titration, the normality of original $\text{K}_2\text{Cr}_2\text{O}_7$ solution is -
(A) 0.3 N (B) 0.2 N (C) 0.6 N (D) 0.1 N

CRX239

Acid Base Titration

69. 20 ml of 0.8 M - H_2SO_4 solution is diluted to 100 ml. 25 ml of the diluted solution exactly require 30 ml NaOH solution for titration. The molarity of NaOH solution is -
(A) $\frac{2}{15}$ M (B) $\frac{4}{15}$ M (C) $\frac{1}{15}$ M (D) $\frac{1}{60}$ M

CRX238

Back Titration

70. 10 g CaCO_3 were dissolved in 250 mL of 1 M HCl. What volume of 2 M KOH would be required to neutralise excess HCl ?
(A) 75 mL (B) 25 mL (C) 15 mL (D) 50 mL
71. 35 ml of 1.5 M - H_2SO_4 solution is mixed with 50 ml of 0.6 M-NaOH solution. If the resulting solution exactly requires 75 ml of KOH solution for the neutralisation of excess acid, the molarity of KOH solution is -
(A) 0.5M (B) 0.75 M (C) 2.0 M (D) 1.0 M

CRX202

CRX240

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Oxidation Number

- In the compound NOClO_4 , the oxidation state of
 (A) nitrogen is +1 (B) nitrogen is +3 (C) chlorine is +5 (D) chlorine is +7
CRX364
- Which of the following is/are peroxide(s)?
 (A) PbO_2 (B) H_2O_2 (C) SrO_2 (D) BaO_2
CRX365
- Which of the following is a non-redox process?
 (A) $\text{SO}_4^{2-} \rightarrow \text{SO}_3$ (B) $\text{Cr}_2\text{O}_7^{2-} \rightarrow \text{CrO}_4^{2-}$ (C) $\text{PO}_4^{3-} \rightarrow \text{P}_2\text{O}_7^{4-}$ (D) $\text{C}_2\text{O}_4^{2-} \rightarrow \text{CO}_2$
CRX366

Balancing of Redox Reaction

- In the reaction, $x\text{Zn} + y\text{HNO}_3(\text{dil}) \longrightarrow a\text{Zn}(\text{NO}_3)_2 + b\text{H}_2\text{O} + c\text{NH}_4\text{NO}_3$, upon balancing with simplest whole number coefficients, select the correct option(s) :
 (A) The sum of the coefficients ($a + b + c$) is 8.
 (B) Whole of HNO_3 is reduced to NH_4^+ in above reaction.
 (C) Zn acts as reductant.
 (D) HNO_3 acts as oxidant.
CRX260
- For the reaction : $\text{I}^- + \text{ClO}_3^- + \text{H}_2\text{SO}_4 \longrightarrow \text{Cl}^- + \text{HSO}_4^- + \text{I}_2$
 The correct statement(s) in the balanced equation is/are :
 (A) Stoichiometric coefficient of HSO_4^- is 6. (B) Iodide is oxidized.
 (C) Sulphur is reduced. (D) H_2O is one of the products.
CRX117

Equivalent Concept

- Which of the following samples of reducing agents is/are chemically equivalent to 25 mL of 0.2 N KMnO_4 to be reduced to Mn^{2+} and water :
 (A) 25 mL of 0.2 M FeSO_4 to be oxidized to Fe^{3+}
 (B) 50 mL of 0.1 M H_3AsO_3 to be oxidized to H_3AsO_4
 (C) 25 mL of 0.1 M H_2O_2 to be oxidized to H^+ and O_2
 (D) 25 mL of 0.1 M SnCl_2 to be oxidized to Sn^{4+}
CRX242
- 100 ml of 0.4 M -acidified KMnO_4 solution may be decolourised completely by
 (A) 200 ml 1N - $\text{K}_2\text{Cr}_2\text{O}_7$ solution (B) 300 ml 0.5M - H_2O_2 solution
 (C) 100 ml 0.8N - KI solution (D) 75 ml 1.4 N - $\text{H}_2\text{C}_2\text{O}_4$ solution
CRX064

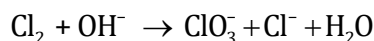
8. The correct information(s) about the n-factor of given species in the reaction :



- (A) $\text{KMnO}_4 \rightarrow 5$ (B) $\text{H}_2\text{SO}_4 \rightarrow 10/3$
(C) $\text{HCl} \rightarrow 1$ (D) $\text{K}_2\text{SO}_4 \rightarrow 2$

CRX367

9. 0.2 mole of Cl_2 is reacted with 500 ml of 1.2 M - NaOH solution



The incorrect information(s) related is/are -

- (A) Cl_2 is limiting reagent (B) 0.4 mole NaOH will left unreacted
(C) $[\text{ClO}_3^-]_{\text{final solution}} = 0.133 \text{ M}$ (D) $[\text{Cl}^-]_{\text{final solution}} = 0.267 \text{ M}$

CRX368

MATRIX MATCH TYPE QUESTION

Oxidation Number

10. Match List-I (Compounds) with List-II (Oxidation states of Nitrogen) and select answer using the codes given below the lists :

List-I

- (a) NaN_3
(b) N_2H_2
(c) NO
(d) N_2O_5
(A) (a)-3; (b)-4; (c)-2; (d)-1
(B) (a)-4; (b)-3; (c)-2; (d)-1
(C) (a)-3; (b)-4; (c)-1; (d)-2
(D) (a)-4; (b)-3; (c)-1; (d)-2

List-II

- (1) +5
(2) +2
(3) -1/3
(4) -1

CRX108

11. Match the column-

Column-I	Column-II
(A) $\text{P}_2\text{H}_4 \longrightarrow \text{PH}_3 + \text{P}_4\text{H}_2$	(p) $E = \frac{3M}{4}$
(B) $\text{I}_2 \longrightarrow \text{I}^\ominus + \text{IO}_3^\ominus$	(q) $E = \frac{3M}{5}$
(C) $\text{MnO}_4^\ominus + \text{Mn}^{2\oplus} + \text{H}_2\text{O} \longrightarrow \text{Mn}_3\text{O}_4 + \text{H}^\oplus$	(r) $E = \frac{15M}{26}$
(D) $\text{H}_3\text{PO}_2 \longrightarrow \text{PH}_3 + \text{H}_3\text{PO}_3$	(s) $E = \frac{5M}{6}$

- (A) A-r; B-q; C-r; d-s (B) A-s; B-q; C-r; d-p
(C) A-p; B-q; C-r; d-s (D) A-s; B-p; C-q; d-r

CRX068

ASSERTION REASON TYPE QUESTION

12. **Statement-1 :-** Moles of KMnO_4 required for oxidation of Fe^{2+} in acidic and basic medium will be different.

Statement-2 :- Final oxidation state to which Mn^{7+} will be reduced will be different in case of acidic and basic medium.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CRX369

13. **Statement-1 :-** Mass of a particular substance that combine with 8 gm of oxygen is said to be equivalent weight of substance.

Statement-2 :- x gm of metal gave y gm of its oxide, so equivalent weight of metal is $\left(\frac{x}{y-x}\right) \times 8$

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CRX370

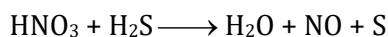
EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

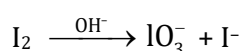
Equivalent weight & n-factor

1. What is the equivalent weight of HNO_3 in following reaction ?



CRX269

2. Find out the n_{factor} of IO_3^- in the following disproportionation reaction.



CRX271

Equivalent Concept

3. Metallic tin in the presence of HCl is oxidized by $\text{K}_2\text{Cr}_2\text{O}_7$ to stannic chloride, SnCl_4 . What volume (in litre) of deci-normal dichromate solution would be reduced by 11.9 gm of tin [$\text{Sn} = 119$]

CRX283

4. Calculate the millimoles of Br_2 produced when 10 ml of 0.1 M BrO_3^- reacts with excess of Br^- .

CRX284

Redox Titration

5. A 458 g sample containing Mn_3O_4 was dissolved and all manganese was converted to Mn^{2+} . In the presence of fluoride ion, Mn^{2+} is titrated with 3 L of KMnO_4 solution (which was 1.25 N against oxalate in acidic medium), both reactants being converted to a complex of Mn(III), What was the % of Mn_3O_4 in the sample ?

CRX247

Acid Base Titration

6. How many millimoles of gaseous ammonia will be required to be passed into 30 cm^3 of 1 N H_2SO_4 solution to bring down the acid strength of the latter to 0.2 N ?

CRX244

7. H_3PO_4 is a tri basic acid and one of its salt is NaH_2PO_4 . What volume (in ml) of 1 M NaOH solution should be added to 12 g of NaH_2PO_4 to convert it into Na_3PO_4 ?

CRX282

Iodimetric and Iodometric Titration

8. 80 g of a sample of anhydrous CuSO_4 was dissolved in water and made to 250 mL. 25 mL of this solution after taking usual precaution was treated with a little excess of KI solution. A white ppt. of CuI and iodine was evolved. The iodine so evolved required 40 mL of 1 M hypo solution. What is the approximate purity of CuSO_4 sample ? [Take mol. wt. of $\text{CuSO}_4 = 160$ u]

CRX246

Redox and Equivalent Concept

9. 1 g sample of AgNO_3 is dissolved in 50 mL of water. It is titrated with 50 mL of KI solution. The AgI precipitated is filtered off. Excess of KI in filtrate is titrated with M/10 KIO_3 in presence of 6M HCl till all I^- converted into ICl. It requires 50 mL of M/10 KIO_3 solution. 20 mL of the same stock solution of KI requires 30 mL of M/10 KIO_3 under similar conditions. Calculate % of AgNO_3 in sample. The reaction is :
 $\text{KIO}_3 + 2\text{KI} + 6\text{HCl} \longrightarrow 3\text{ICl} + 3\text{KCl} + 3\text{H}_2\text{O}$
 CRX248
10. What volume (in ml) of 0.5 M- $\text{Na}_2\text{S}_2\text{O}_3$ solution will react completely with 12.7 gm iodine ?
 (I = 127)
 CRX285

Double Titration

11. How many litre of 0.1N HCl are required to react completely with 19 gm mixture of Na_2CO_3 and NaHCO_3 containing equimolar amount of these two ?
 CRX281

Hardness of water

12. 50 L of a water sample contains 7.3 gm $\text{Mg}(\text{HCO}_3)_2$. Calculate temporary hardness of the water in terms of ppm of CaCO_3 .
 CRX298

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Oxidation and Reduction

13. Classify each of the following unbalanced half-reactions as either an oxidation or a reduction :
 (a) $\text{O}_2(\text{g}) \longrightarrow \text{OH}^-(\text{aq})$ (b) $\text{H}_2\text{O}_2(\text{aq}) \longrightarrow \text{O}_2(\text{g})$
 (c) $\text{MnO}_4^- (\text{aq}) \longrightarrow \text{MnO}_4^{2-}(\text{aq})$ (d) $\text{CH}_3\text{OH}(\text{aq}) \longrightarrow \text{CH}_2\text{O}(\text{aq})$
 (e) $\text{MnO}_2(\text{s}) \longrightarrow \text{Mn}_3\text{O}_4(\text{s})$ (f) $\text{FeO}(\text{s}) \longrightarrow \text{Fe}_3\text{O}_4(\text{s})$
 (g) $\text{CH}_3\text{COOH}(\text{aq.}) \longrightarrow \text{CH}_3\text{CH}_2\text{OH}(\text{aq})$
 CRX266

2. Oxidation Number

14. Calculate oxidation number of the underlined elements in the following compounds or ions

- | | | | |
|--|--|---|--|
| (a) $\text{H}_4\underline{\text{P}}\text{O}_7$ | (b) $\underline{\text{S}}_2\text{O}_3^{2-}$ | (c) $\text{Ca}\underline{\text{O}}_2$ | (d) $\underline{\text{Zn}}\text{O}_2^{2-}$ |
| (e) $\underline{\text{Cr}}\text{O}_2\text{Cl}_2$ | (f) $\text{Ca}(\underline{\text{Cl}}\text{O}_2)_2$ | (g) $\underline{\text{C}}_3\text{O}_2$ | (h) $\text{K}\underline{\text{I}}_3$ |
| (i) $\underline{\text{Fe}}_3\text{O}_4$ | (j) $\underline{\text{Br}}_3\text{O}_8$ | (k) $\text{CaO}\underline{\text{Cl}}_2$ | (l) $\underline{\text{O}}_2^+$ |
| (m) $\underline{\text{O}}_2^-$ | (n) $\underline{\text{C}}\text{N}^-$ | | |

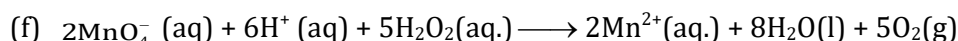
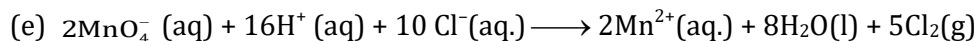
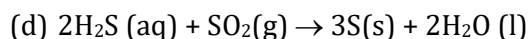
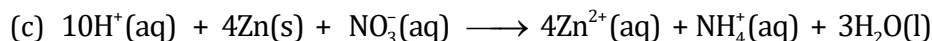
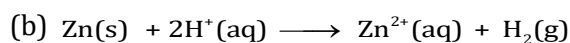
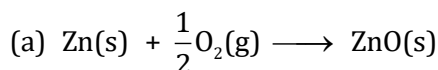
CRX264

15. Calculate oxidation number of the underlined elements in the following compounds or ions

- | | |
|--|--|
| (a) $\text{K}[\underline{\text{Co}}(\text{C}_2\text{O}_4)_2(\text{NH}_3)_2]$ | (b) $\text{K}\underline{\text{Al}}(\underline{\text{S}}\text{O}_4)_2 \cdot 12\text{H}_2\text{O}$ |
| (c) $[\underline{\text{Fe}}(\text{CN})_6]^{4-}$ | (d) $\underline{\text{Fe}}_{0.93}\text{O}$ |
| (e) $\text{H}_2\underline{\text{S}}_2\text{O}_8$ | (f) $\underline{\text{Cr}}\text{O}_5$ |
| (g) $\text{H}_2\underline{\text{S}}\text{O}_5$ | |

CRX265

16. Identify the oxidant (oxidising agent) and reductant (reducing agent) in the following reactions



CRX267

Equivalent weight & n-factor

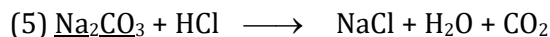
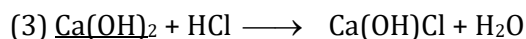
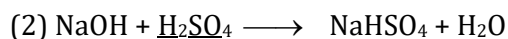
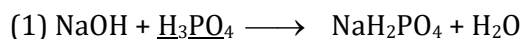
17. 1.12 litre dry chlorine gas at NTP (1 ATM & 273K) was passed over a heated metal when 5.56 g of chloride of the metal was formed. What is the equivalent weight of the metal ?

CRX204

18. 0.98 g of the metal sulphate was dissolved in water and excess of barium chloride was added. The precipitated barium sulphate weighed 0.95 g. Calculate the equivalent weight of the metal.

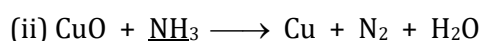
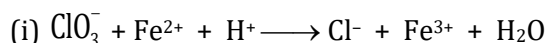
CRX205

19. Find the n-factor of underlined species in the following non redox reaction.



CRX268

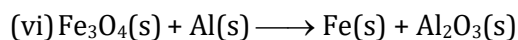
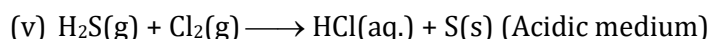
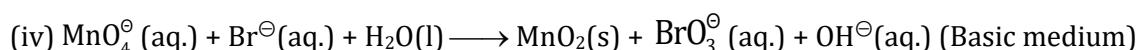
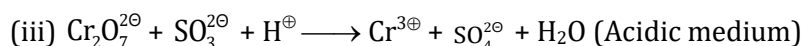
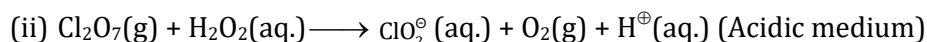
20. Find out the equivalent weight of the underlined species in the following reaction :



CRX270

Balancing of Redox Reactions by Oxidation Number Method

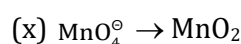
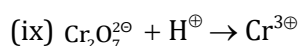
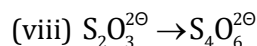
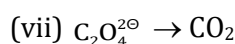
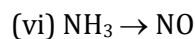
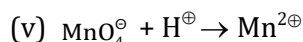
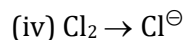
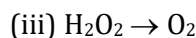
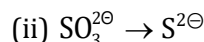
21. Balance the following redox reactions by oxidation number method :



CRX262

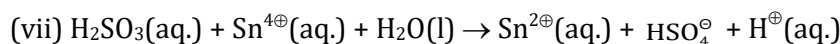
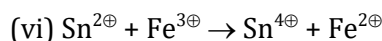
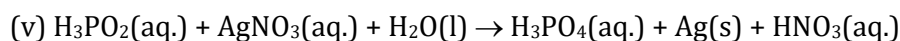
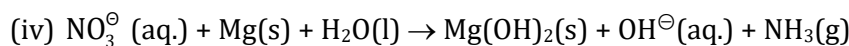
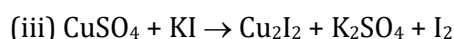
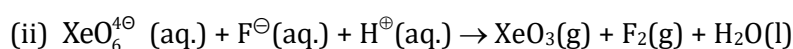
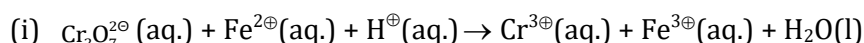
Balancing of Redox Reactions by Ion Electron Method

22. Balance following half reaction in acidic medium



CRX261

23. Balance the following redox reactions by half-reaction method (Ion electron method):



CRX263

Acid Base Titration24. Calculate volume (in L) of 0.4 M NaOH required to react with following mixture
 $\text{HCl}(1 \text{ mol}) + \text{H}_2\text{SO}_4(2 \text{ mol})$

CRX280

25. 4.00 gm NaOH is dissolved in 100 ml HCl solution and the solution is diluted to 500 ml. If 25 ml of the diluted solution exactly requires 10 ml of 0.02 M Na_2CO_3 solution for titration, calculate the concentration of original HCl solution (in gm/litre).

CRX288

Iodimetric and Iodometric Titration26. 5 g of pyrolusite (impure MnO_2) were heated with conc. HCl and Cl_2 evolved was passed through excess of KI solution. The iodine liberated required 40 mL of N/10 hypo solution. Find the % of MnO_2 in the pyrolusite.

CRX245

27. 25 ml of 0.4 M -acidified KMnO_4 solution is treated with excess of KI solution. The liberated I_2 exactly requires 20 ml of $\text{Na}_2\text{S}_2\text{O}_3$ solution for titration. What is the molarity of $\text{Na}_2\text{S}_2\text{O}_3$ solution?

CRX287

Back Titration28. 20 ml of 2M H_2SO_4 solution is mixed with 30 ml of 3M NaOH solution. Calculate the volume (in ml) of 4M-HCl solution needed to neutralise the resulting solution.

CRX286

Hardness of water29. A sample of hard water contains 324 ppm of $\text{Ca}(\text{HCO}_3)_2$. How many gm of CaO is needed to remove all bicarbonates from 100 L hard water?

CRX297

EXERCISE - JEE (Main) PYQ

- From the given list, the number of compounds with + 4 oxidation state of Sulphur _____.
 SO_3 , H_2SO_3 , SOCl_2 , SF_4 , BaSO_4 , $\text{H}_2\text{S}_2\text{O}_7$ [JEE (Main) 2024]
 CRX371
- Which of the following cannot function as an oxidising agent? [JEE (Main) 2024]
 (1) N^{3-} (2) SO_4^{2-} (3) BrO_3^- (4) MnO_4^-
 CRX372
- Match the **List-I** with **List-II**

List-I (Redox Reaction)		List-II (Type of Redox Reaction)	
A	$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \xrightarrow{\Delta} \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	(I)	Disproportionation reaction
B	$2\text{NaH}(\text{s}) \xrightarrow{\Delta} 2\text{Na}(\text{s}) + \text{H}_2(\text{g})$	(II)	Combination reaction
C	$\text{V}_2\text{O}_5(\text{s}) + 5\text{Ca}(\text{s}) \xrightarrow{\Delta} 2\text{V}(\text{s}) + 5\text{CaO}(\text{s})$	(III)	Decomposition reaction
D	$2\text{H}_2\text{O}_2(\text{aq}) \xrightarrow{\Delta} 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$	(IV)	Displacement reaction

Choose the **correct** answer from the options given below : [JEE (Main) 2025]
 (1) A-II, B-III, C-IV, D-I (2) A-II, B-III, C-I, D-IV
 (3) A-III, B-IV, C-I, D-II (4) A-IV, B-I, C-II, D-III
 CRX383
- Which of the following reactions are disproportionation reactions?
 (A) $\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$
 (B) $3\text{MnO}_4^{2-} + 4\text{H}^+ \rightarrow 2\text{MnO}_4^- + \text{MnO}_2 + 2\text{H}_2\text{O}$
 (C) $2\text{KMnO}_4 \rightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$
 (D) $2\text{MnO}_4^- + 3\text{Mn}^{2+} + 2\text{H}_2\text{O} \rightarrow 5\text{MnO}_2 + 4\text{H}^+$

Choose the correct answer from the options given below: [JEE (Main) 2024]
 (1) (A), (B) (2) (B), (C), (D) (3) (A), (B), (C) (4) (A), (D)
 CRX373
- Which of the given reactions is not an example of disproportionation reaction ? [JEE (Main) 2022]
 (1) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ (2) $2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_3 + \text{HNO}_2$
 (3) $\text{MnO}_4^\ominus + 4\text{H}^\oplus + 3\text{e}^\ominus \rightarrow \text{MnO}_2 + 2\text{H}_2\text{O}$ (4) $3\text{MnO}_4^{2\ominus} + 4\text{H}^\oplus \rightarrow 2\text{MnO}_4^\ominus + \text{MnO}_2 + 2\text{H}_2\text{O}$
 CRX049
- An example of a disproportionation reaction is : [JEE (Main) 2019]
 (1) $2\text{KMnO}_4 \rightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$
 (2) $2\text{MnO}_4^\ominus + 10\text{I}^\ominus + 16\text{H}^\oplus \rightarrow 2\text{Mn}^{2\oplus} + 5\text{I}_2 + 8\text{H}_2\text{O}$
 (3) $2\text{CuBr} \rightarrow \text{CuBr}_2 + \text{Cu}$
 (4) $2\text{NaBr} + \text{Cl}_2 \rightarrow 2\text{NaCl} + \text{Br}_2$
 CRX035
- 1 mole of PbS is oxidised by "X" moles of O_3 to get "Y" moles of O_2 . $X + Y =$ _____ [JEE (Main) 2024]
 CRX374

8. Number of moles of H^+ ions required by 1 mole of MnO_4^- to oxidise oxalate ion to CO_2 is ____.
[JEE (Main) 2024]
CRX375
9. Chlorine undergoes disproportionation in alkaline medium as shown below :
 $aCl_2(g) + bOH^-(aq) \longrightarrow cClO^-(aq) + dCl^-(aq) + eH_2O(l)$
 The values of a, b, c and d in a balanced redox reaction are respectively : [JEE (Main) 2024]
 (1) 1, 2, 1 and 1 (2) 2, 2, 1 and 3
 (3) 3, 4, 4 and 2 (4) 2, 4, 1 and 3
 CRX376
10. $2MnO_4^- + bI^- + cH_2O \rightarrow xI_2 + yMnO_2 + zOH^-$
 If the above equation is balanced with integer coefficients, the value of z is ____
 [JEE (Main) 2024]
 CRX377
11. Given below are two statements :
 Statement I: S_8 solid undergoes disproportionation reaction under alkaline conditions to form S^{2-} and $S_2O_3^{2-}$
 Statement II: ClO_4^- can undergo disproportionation reaction under acidic condition.
 In the light of the above statements, choose the most appropriate answer from the options given below : [JEE (Main) 2024]
 (1) Statement I is correct but statement II is incorrect.
 (2) Statement I is incorrect but statement II is correct
 (3) Both statement I and statement II are incorrect
 (4) Both statement I and statement II are correct
 CRX378
12. In acidic medium, $K_2Cr_2O_7$ shows oxidising action as represented in the half reaction
 $Cr_2O_7^{2-} + xH^+ + Ye^- \rightarrow 2A + ZH_2O$
 X, Y, Z and A are respectively are : [JEE (Main) 2024]
 (1) 8, 6, 4 and Cr_2O_3 (2) 14, 7, 6 and Cr^{3+}
 (3) 8, 4, 6 and Cr_2O_3 (4) 14, 6, 7 and Cr^{3+}
 CRX379
13. Some CO_2 gas was kept in a sealed container at a pressure of 1 atm and at 273 K. This entire amount of CO_2 gas was later passed through an aqueous solution of $Ca(OH)_2$. The excess unreacted $Ca(OH)_2$ was later neutralized with 0.1 M of 40 mL HCl. If the volume of the sealed container of CO_2 was x, then x is ____ cm^3 (nearest integer).
 [Given : The entire amount of $CO_2(g)$ reacted with exactly half the initial amount of $Ca(OH)_2$ present in the aqueous solution.] [JEE (Main) 2025]
 CRX384
14. A 100 mL solution was made by adding 1.43 g of $Na_2CO_3 \cdot xH_2O$. The normality of the solution is 0.1 N. The value of x is ____.
 (The atomic mass of Na is 23g/mol) :- [JEE (Main) 2020]
 CRX039
15. In order to oxidise a mixture one mole of each of FeC_2O_4 , $Fe_2(C_2O_4)_3$, $FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium, the number of moles of $KMnO_4$ required is - [JEE (Main) 2019]
 (1) 3 (2) 2 (3) 1 (4) 1.5
 CRX037

16. The strength of an aqueous NaOH solution is most accurately determined by titrating :
(Note : consider that an appropriate indicator is used) [JEE (Main) 2020]
(1) Aq. NaOH in a volumetric flask and concentrated H_2SO_4 in a conical flask
(2) Aq. NaOH in a pipette and aqueous oxalic acid in a burette
(3) Aq. NaOH in a burette and concentrated H_2SO_4 in a conical flask
(4) Aq. NaOH in a burette and aqueous oxalic acid in a conical flask
CRX038
17. Only 2 mL of KMnO_4 solution of unknown molarity is required to reach the end point of a titration of 20 mL of oxalic acid (2 M) in acidic medium. The molarity of KMnO_4 solution should be ____ M.
[JEE (Main) 2024]
CRX380
18. If 50 mL of 0.5 M oxalic acid is required to neutralise 25 mL of NaOH solution, the amount of NaOH in 50 mL of given NaOH solution is ____ g.
[JEE (Main) 2024]
CRX381
19. The density of a monobasic strong acid (Molar mass 24.2 g mol) is 1.21 kg L. The volume of its solution required for the complete neutralization of 25 mL of 0.24 M NaOH is ____ $\times 10^{-2}$ mL (Nearest integer)
[JEE (Main) 2023]
CRX309
20. 20 mL of calcium hydroxide was consumed when it was reacted with 10 mL of unknown solution of H_2SO_4 . Also 20 mL standard solution of 0.5 M HCl containing 2 drops of phenolphthalein was titrated with calcium hydroxide the mixture showed pink colour when burette displayed the value of 35.5 mL whereas the burette showed 25.5 mL initially. The concentration of H_2SO_4 is ____ M (Nearest integer)
[JEE (Main) 2023]
CRX310
21. 50 mL of 0.5 M oxalic acid is needed to neutralize 25 mL of sodium hydroxide solution. The amount of NaOH in 50 mL of the given sodium hydroxide solution is :
[JEE (Main) 2019]
(1) 4 g (2) 2 g (3) 8 g (4) 1 g
CRX036
22. 0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solution. According to equation
 $5\text{I}^- + \text{IO}_3^- + 6\text{H}^+ \rightarrow 3\text{I}_2 + 3\text{H}_2\text{O}$
Identify the **correct** statements :
(A) 200 mL of KI solution reacts with 0.004 mol of KIO_3
(B) 200 mL of KI solution reacts with 0.006 mol of H_2SO_4
(C) 0.5 L of KI solution produced 0.005 mol of I_2
(D) Equivalent weight of KIO_3 is equal to $\left(\frac{\text{Molecular weight}}{5}\right)$
Choose the **correct** answer from the options given below: [JEE (Main) 2025]
(1) (A) and (D) only (2) (B) and (C) only (3) (A) and (B) only (4) (C) and (D) only
CRX385
23. Thiosulphate reacts differently with iodine and bromine in the reaction given below :
 $2\text{S}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$
 $\text{S}_2\text{O}_3^{2-} + 5\text{Br}_2 + 5\text{H}_2\text{O} \rightarrow 2\text{SO}_4^{2-} + 4\text{Br}^- + 10\text{H}^+$
Which of the following statement justifies the above dual behaviour of thiosulphate?
(1) Bromine undergoes oxidation and iodine undergoes reduction by iodine in these reactions
(2) Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reaction
(3) Bromine is a stronger oxidant than iodine
(4) Bromine is a weaker oxidant than iodine
[JEE (Main) 2024]
CRX382

EXERCISE - JEE (Advanced) PYQ

1. The order of the oxidation state of the phosphorus atom in H_3PO_2 , H_3PO_4 , H_3PO_3 and $\text{H}_4\text{P}_2\text{O}_6$ is
[JEE (Advanced) 2017]
(A) $\text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2$
(B) $\text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6$
(C) $\text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_4$
(D) $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6$
CRX076
2. Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen-
[JEE (Advanced) 2012]
(A) HNO_3 , NO, NH_4Cl , N_2 (B) HNO_3 , NO, N_2 , NH_4Cl
(C) HNO_3 , NH_4Cl , NO, N_2 (D) NO, HNO_3 , NH_4Cl , N_2
CRX071
3. In neutral or faintly alkaline solution, 8 moles permanganate anion quantitatively oxidize thiosulphate anions to produce X moles of a sulphur containing product. the magnitude of X is
[JEE (Advanced) 2016]
CRX075
4. For the reaction
 $\text{I}^\ominus + \text{ClO}_3^\ominus + \text{H}_2\text{SO}_4 \rightarrow \text{Cl}^\ominus + \text{HSO}_4^\ominus + \text{I}_2$
The correct statement(s) in the balanced equation is / are :
[JEE (Advanced) 2014]
(A) Stoichiometric coefficient of $\text{HSO}_4^\ominus$ is 6 (B) Iodide is oxidized
(C) Sulphur is reduced (D) H_2O is one of the products
CRX073
5. The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc. HNO_3 to a compound with the highest oxidation state of sulphur is ____
(Given data : Molar mass of water = 18 g mol^{-1})
[JEE (Advanced) 2019]
CRX080
6. In the chemical reaction between stoichiometric quantities of KMnO_4 and KI in weakly basic solution, what is the number of moles of I_2 released for 4 moles of KMnO_4 consumed ?
[JEE (Advanced) 2020]
CRX083
7. H_2S (5 moles) reacts completely with acidified aqueous potassium permanganate solution. In this reaction, the number of moles of water produced is x, and the number of moles of electrons involved is y. The value of (x + y) is ____.
[JEE (Advanced) 2023]
CRX313

Question Stem for Question Nos. 8 and 9

A sample (5.6 g) containing iron is completely dissolved in cold dilute HCl to prepare a 250 mL of solution. Titration of 25.0 mL of this solution requires 12.5 mL of 0.03 M KMnO_4 solution to reach the end point. Number of moles of Fe^{2+} present in 250 mL solution is $x \times 10^{-2}$ (consider complete dissolution of FeCl_2). The amount of iron present in the sample is y% by weight.

(Assume : KMnO_4 reacts only with Fe^{2+} in the solution)

Use : Molar mass of iron as 56 g mol^{-1})

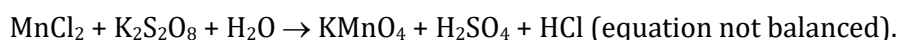
8. The value of x is _____. [JEE (Advanced) 2021]

CRX084

9. The value of y is _____. [JEE (Advanced) 2021]

CRX085

10. To measure the quantity of MnCl_2 dissolved in an aqueous solution, it was completely converted to KMnO_4 using the reaction,



Few drops of concentrated HCl were added to this solution and gently warmed. Further, oxalic acid (225 mg) was added in portions till the colour of the permanganate ion disappeared. The quantity of MnCl_2 (in mg) present in the initial solution is ____.

[JEE (Advanced) 2018]

(Atomic weights in g mol^{-1} : Mn = 55, Cl = 35.5)

CRX079

11. 5.00 mL of 0.10 M oxalic acid solution taken in a conical flask is titrated against NaOH from a burette using phenolphthalein indicator. The volume of NaOH required for the appearance of permanent faint pink color is tabulated below for five experiments. What is the concentration, in molarity, of the NaOH solution ? [JEE (Advanced) 2020]

Exp. No.	Vol. of NaOH (ml)
1	12.5
2	10.5
3	9
4	9
5	9

CRX082

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	B	D	C	D	D	D	C	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	C	A	C	D	A	A	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	D	D	B	D	C	C	B	B	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	C	B	A	D	C	C	B	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	A	C	A	B	A	D	C	D	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	C	A	D	C	D	D	D	A	D	B
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	D	A	B	B	A	C	A	C	B	B
Que.	71									
Ans.	D									

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	BD	BCD	ABC	ACD	ABD	ACD	B	ABC	BD	A
Que.	11	12	13							
Ans.	B	A	B							

EXERCISE (S)

1. 21 2. 5 3. 4 4. 3 5. 50
6. 24 7. 200 8. 80 9. 85 10. 200
11. 3.00 12. 100
13. (a) Reduction, (b) Oxidation, (c) Reduction, (d) Oxidation, (e) Reduction, (f) Oxidation, (g) Reduction
14. (a) +5, (b) +2, (c) -1, (d) +2, (e) +6, (f) +3, (g) +4/3, (h) -1/3, (i) +8/3, (j) +16/3, (k) 0, (l) +1/2, (m) -1/2, (n) +2
15. (a) +3 (b) +3, +6 (c) +2 (d) $+\frac{200}{93}$ (e) +6 (f) +6 (g) +6
16. (a) Oxidant : O_2 ; Reductant : Zn (b) Oxidant : H^+ ; Reductant : Zn
(c) Oxidant : NO_3^- ; Reductant : Zn (d) Oxidant : SO_2 ; Reductant : H_2S
(e) Oxidant : MnO_4^- ; Reductant : Cl^- (f) Oxidant : MnO_4^- ; Reductant : H_2O_2
17. 20.1 18. 72.61 19. (1) 1 (2) 1 (3) 1 (4) 1 (5) 2
20. (i) 13.91 or 13.92, (ii) 5.66 or 5.67 21. Theoretical 22. Theoretical
23. Theoretical 24. 12.50 25. 39.42 26. 3.48
27. 2.50 28. 2.50 29. 11.20

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	1	1	3	3	8	8	1	8
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	45	10	2	4	8	4	12	1
Que.	21	22	23							
Ans.	1	1	3							

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	6	ABD	288	6	18	1.875	18.75	126
Que.	11									
Ans.	0.11									